

Assignment No 17

Name : Shivam Ramakant Deshmane

Domain : AI & ML

College : T.B Girwalkar Polytechnic Ambajogai

Q1.

First 10 Rows:

```
...   age  sex  cp  trestbps  chol  fbs  restecg  thalach  exang  oldpeak  slope
0    63    1   3    145    233    1         0    150      0      2.3      0
1    37    1   2    130    250    0         1    187      0      3.5      0
2    41    0   1    130    204    0         0    172      0      1.4      2
3    56    1   1    120    236    0         1    178      0      0.8      2
4    57    0   0    120    354    0         1    163      1      0.6      2
5    57    1   0    140    192    0         1    148      0      0.4      1
6    56    0   1    140    294    0         0    153      0      1.3      1
7    44    1   1    120    263    0         1    173      0      0.0      2
8    52    1   2    172    199    1         1    162      0      0.5      2
9    57    1   2    150    168    0         1    174      0      1.6      2
```

```
   ca  thal  target
0    0     1       1
1    0     2       1
2    0     2       1
3    0     2       1
4    0     2       1
5    0     1       1
6    0     2       1
7    0     3       1
8    0     3       1
9    0     2       1
```

Shape of the Dataset:

(303, 14)

Basic Information:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 303 entries, 0 to 302

Data columns (total 14 columns):

#	Column	Non-Null Count	Dtype
0	age	303 non-null	int64
1	sex	303 non-null	int64
2	cp	303 non-null	int64
3	trestbps	303 non-null	int64
4	chol	303 non-null	int64
5	fbs	303 non-null	int64
6	restecg	303 non-null	int64
7	thalach	303 non-null	int64
8	exang	303 non-null	int64
9	oldpeak	303 non-null	float64
10	slope	303 non-null	int64
11	ca	303 non-null	int64
12	thal	303 non-null	int64
13	target	303 non-null	int64

dtypes: float64(1), int64(13)

Q2.

... Missing values in each column:

age	0
sex	0
cp	0
trestbps	0
chol	0
fbs	0
restecg	0
thalach	0
exang	0
oldpeak	0
slope	0
ca	0
thal	0
target	0
dtype:	int64

Q3.



... Number of duplicate rows:
1

New Shape of the Dataset:
(302, 14)

Q4.



```
... Number of rows with Cholesterol = 0: 0  
Number of rows with RestingBP = 0: 147
```

Q5.

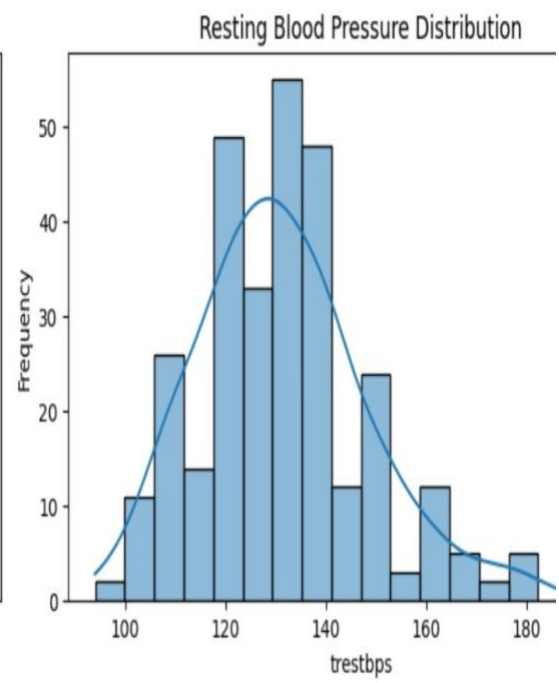
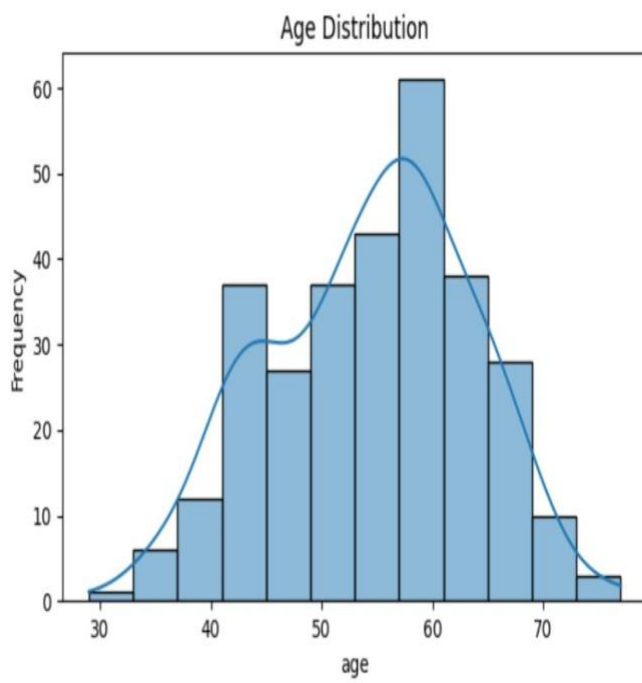
✓

```
... Before Cleaning:
      chol      restecg
count 303.000000 303.000000
mean  246.264026  0.528053
std    51.830751  0.525860
min    126.000000  0.000000
25%    211.000000  0.000000
50%    240.000000  1.000000
75%    274.500000  1.000000
max     564.000000  2.000000

After Cleaning:
      chol      restecg
count 303.000000 303.000000
mean  246.264026  1.027756
std    51.830751  0.113623
min    126.000000  1.000000
25%    211.000000  1.000000
50%    240.000000  1.000000
75%    274.500000  1.030000
max     564.000000  2.000000
```

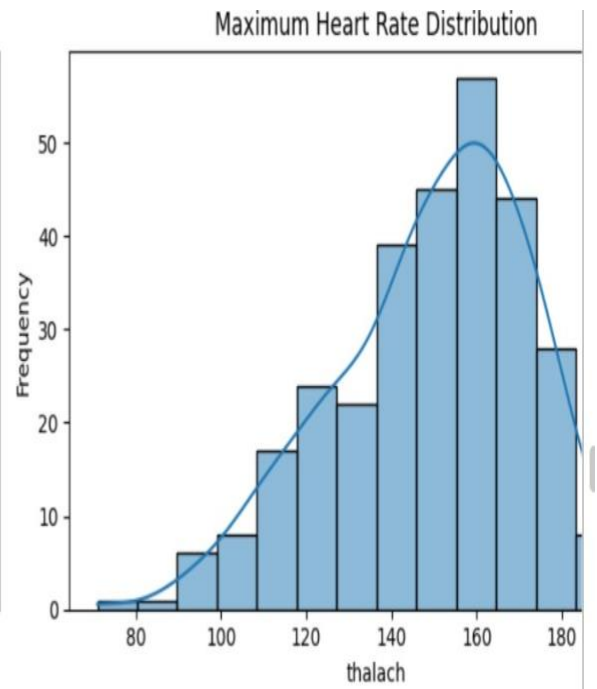
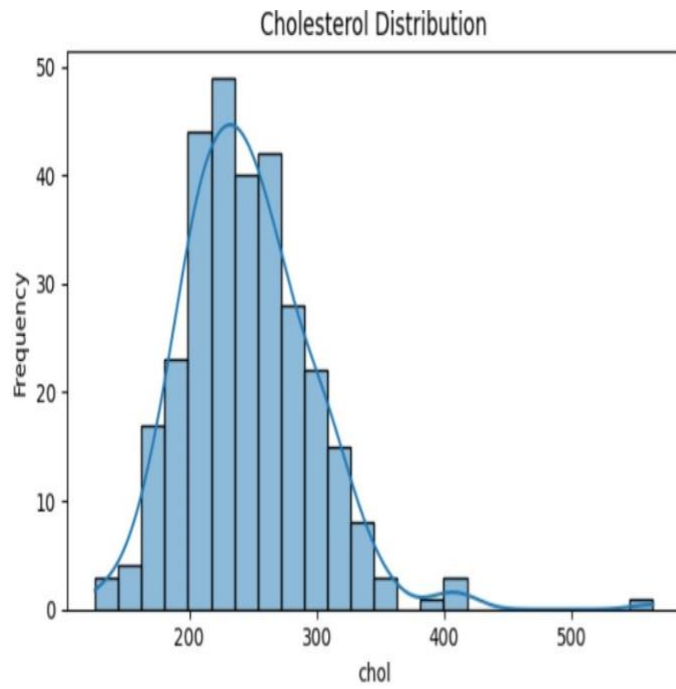
Q6.

...



▼

...



Q7.

```
▼ *** Numerical Columns:
    ['age', 'sex', 'cp', 'trestbps', 'chol', 'fbs', 'restecg', 'thalach', 'exang', 'oldpeak', 'slope', 'ca', 'thal', 'target']

    Categorical Columns:
    []
```

Q8.

▼ ... Shape of Encoded DataFrame:
(303, 14)

First 5 Rows of Encoded DataFrame:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	\
0	63	1	3	145	233	1	0	150	0	2.3	0	
1	37	1	2	130	250	0	1	187	0	3.5	0	
2	41	0	1	130	204	0	0	172	0	1.4	2	
3	56	1	1	120	236	0	1	178	0	0.8	2	
4	57	0	0	120	354	0	1	163	1	0.6	2	

	ca	thal	target
0	0	1	1
1	0	2	1
2	0	2	1
3	0	2	1
4	0	2	1

Q9.

```
v ... Final Shape of df_encoded:  
(303, 27)  
  
Column Names:  
['age', 'trestops', 'chol', 'thalach', 'oldpeak', 'ca', 'target', 'sex_0', 'sex_1', 'cp_0', 'cp_1', 'cp_2', 'cp_3', 'fbs_0', 'fbs_1', 'restecg']
```

Q10.

[8]
' Os



```
# ===== Summary =====

# 1. Invalid Values Found and Fixed:
#   - Some rows had Cholesterol = 0 and RestingBP = 0, which are unrealistic values.
#   - These values were replaced with the mean of the respective columns
#     (excluding zero values), and the columns were rounded to 2 decimal places.

# 2. Importance of Handling Invalid Values:
#   - Invalid values such as Cholesterol = 0 and RestingBP = 0 can reduce
#     the accuracy of data analysis and machine learning models.
#   - Replacing them with appropriate values improves data quality and reliability.

# 3. Purpose of One-Hot Encoding:
#   - One-Hot Encoding converts categorical variables into numerical columns.
#   - This allows machine learning algorithms to process categorical data
#     without assuming any order between categories.

# 4. Other Observations:
#   - The dataset contained no missing values.
#   - Duplicate rows were checked and removed if present.
#   - Numerical features showed different value ranges, making visualization useful.
#   - The dataset is now cleaned, encoded, and ready for further analysis or
#     machine learning model building.
```